# Ocean Sound Separation with Causal High-Resolution Background Priors

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***Abstract*—We test whether past ocean soundscape can predict a background prior that improves separation of a future event. DCLDE examples combine an annotation-clean real 5-s ocean recording with a synthetic marine event; the preceding 60-s history masks annotated calls. An initial 30-band GRU forecast injected through FiLM does not improve BS-RoFormer on 4,096 held-out mixtures (9.497 versus 9.631 dB target SI-SDRi), even with the true current 30-band spectrum as an Oracle (9.535 dB). An information ladder instead identifies a useful 128-group, time-resolved representation. We feed its background map and explicit mixture-minus-background innovation through a matched residual adapter. The true-background Oracle reaches 16.767 dB; fixed-mean and wrong-day map controls reach 12.156 and 11.944 dB, while past-60-s persistence and a frozen GRU forecast reach 14.295 and 14.351 dB. Persistence exceeds the fixed-mean and wrong-day controls by 2.138 and 2.351 dB, respectively. Forecast exceeds persistence by $0.056$ dB (recording-day paired 95% CI $[0.036, 0.074]$), but remains $2.416$ dB below Oracle. Thus the residual interface itself helps, and time-relevant historical background supplies a further substantial gain; this GRU adds little beyond persistence. Synthetic targets and only three held-out days limit generalization; classifier tests on untouched calls do not establish real-event superiority. On the same 103 real calls, causal conditions preserve off-event audio better than wrong-day Shuffled, whereas Shuffled scores better on the classifier's two event-probability measures; the real-event ranking is therefore unresolved. Training-day empirical targets reduce equal-size real-to-target embedding distance from $.483$ to $.322$. Retraining with this 50/50 procedural/pseudo-real pool lowers $p(\hat{N})$ from $.829$ to $.652$ for Persistence on the same real calls, while off-event change rises from $.037$ to $.058$ and old-synthetic SI-SDRi falls from 14.295 to 12.195 dB. Target realism improves a semantic proxy but introduces a measurable preservation and distribution tradeoff. A separate framewise detector enables train-day-only weak fine-tuning on annotated real calls: held-out $p(\hat{N})$ falls further to $.408$ and $p(\hat{s})$ reaches $.978$, while old-synthetic SI-SDRi declines to 10.323 dB. Independent evaluation and off-event checks support a partial real-domain gain, not full-reference whale-stem recovery. Freezing the separator representation and distilling from the 50/50 teacher retains 12.156 dB on the old synthetic test, with real $p(\hat{N})/p(\hat{s}) = .461/.960$; the stronger real suppression and near-preserved controlled separation remain a tradeoff.**

***Index Terms*—ocean acoustics, source separation, soundscape forecasting, BS-RoFormer, weak-reference evaluation**

## I. Introduction

Ocean backgrounds vary with shipping, weather, biology, and season [1]. A mixture-only separator must infer both an unexpected event and this changing background from the same observation. We instead ask whether the preceding soundscape can supply a causal prior. Our supervised data tuple and main model are

$$\begin{aligned} x_t &= N_t + s_t, \qquad \widehat{B}_t = P_\psi(H_t), \\ \widehat{R}_t &= \log|X_t| - \widehat{B}_t, \\ [X_t, \widehat{B}_t, \widehat{R}_t] &\xrightarrow{\text{Residual+BSR}} (\hat{s}_t, \hat{N}_t), \end{aligned} \tag{1}$$

where $H_t$ excludes annotated call intervals and $N_t$ is a real current no-call segment. Training has exact stems, while the separator never receives $N_t$ or $s_t$ as input. Here $X_t$ denotes the current mixture STFT and $\widehat{B}_t$ its predicted background log-magnitude map. The 30-band FiLM precursor failed on held-out audio, motivating an information ladder and the 128-group residual interface in Fig. 1.

Time-domain separators [2], [3], attention models [4], [5], and band-split systems [6], [7] provide strong mixture-only backbones. Hybrid Transformers and MixIT broaden the architecture and supervision regimes [8], [9], but do not test a forecast of the upcoming background. FiLM, cross-attention, and cross-modal separation [10]–[12] motivate three interfaces. Forecasting baselines range from recurrent to linear and patch-based models [13]–[16].

Our contributions are: (1) a 128-group time-resolved causal background map with an explicit innovation and residual BSR interface;[1] (2) an annotation-aware construction of real clean ocean backgrounds and event-masked 60-s histories; (3) a six-way matched test separating adapter capacity, wrong-day background, persistence, learned forecast, and current Oracle information; and (4) a diagnostic chain from failed 30-band FiLM to the frequency-resolution ladder. Broad source and real-event studies assess coverage and transfer; they are secondary to the full-reference DCLDE test.

## II. Method

### A. High-resolution causal residual separator

The main history $H_t \in \mathbb{R}^{12\times 79\times 128}$ contains twelve strictly prior 5-s, 128-group log-magnitude maps. A 512-point Hann STFT with 128-point hop gives 79 frames per window at 2 kHz. Annotated event windows are masked and filled only with nearby clean windows within the past 60 s. A small residual GRU compresses each map into four temporal chunks, processes the twelve steps, and predicts the next full $79 \times 128$ map $\widehat{B}_t$; its zero-initialized decoder starts at last-map

[1]Code and reproduction instructions: https://github.com/D-eval/ocean_noise_pred_splite.

persistence. The forecaster is trained on clean training days, selected by validation map MAE, and frozen before separation.

We form $\widehat{R}_t = \log|X_t| - \widehat{B}_t$ on the same grid. The time-aligned pair $[\widehat{B}_t, \widehat{R}_t]$ is projected into mixture-derived BSR band tokens before axial attention. The residual projection starts at zero, so every conditioned separator initially equals the same transferred mixture-only BSR waveform mapping. We train with exact synthetic event stems and a waveform plus multi-resolution STFT loss. Figure 1 shows the data and inference paths.

### B. Preliminary 30-band FiLM design

The precursor history $H_t \in \mathbb{R}^{12\times 30}$ contains 30-band log-power features computed on twelve preceding 5-s windows; annotated calls and their 2-s guards are excluded. Features are normalized using training data only. A residual GRU predicts

$$z_t = \mathrm{LN}(\mathrm{GRU}(H_t)), \quad \widehat{S}_t = S_{t-1} + g_\psi(z_t). \tag{2}$$

Residual initialization represents persistence exactly; a learned predictor is retained only if validation MAE improves upon persistence [17].

The precursor FC-BSR maps the mixture STFT to band tokens $X_k$ and applies FiLM before the axial Transformer:

$$\widetilde{X}_k = (1 + \tanh\gamma_k) \odot X_k + \beta_k. \tag{3}$$

The separator predicts an event stem; the ocean stem is the mixture residual. Training combines waveform $L_1$ and multi-resolution complex-STFT losses. Concatenation and cross-attention were also tested as preliminary interfaces.

The preliminary 30-band audit holds the separator backbone, mixtures, initialization, and training rule fixed. A true BS-RoFormer bypasses the condition injector; Zero-FiLM retains the FiLM branch and the same frozen GRU, but replaces its output by $c = 0$. Shuffled-FiLM uses $E(N_j)$ from a deterministic half-cycle offset in the clean-window index of the same split ($j \neq t$); on the held-out test, this selects a different day. Persistence-FiLM uses $S_{t-1}$, and Forecast-FiLM uses the frozen GRU output. Oracle-FiLM receives $S_t^{\mathrm{oracle}} = E(N_t)$ computed from the actual current clean background after any shared mixture scaling. We call this the *30-band Oracle-FiLM*: $E$ removes temporal structure and phase, so this row bounds neither a full-background condition nor the value of perfect forecasting with a richer representation. It is available only in the controlled experiment.

### C. Marine-aware source distribution

The broad generator contains 13 families: harmonic, linear chirp, inharmonic, impulse train, frequency-modulated tone, band noise, whale upsweep, downsweep, arch, S-curve, pulse packet, vessel tonal, and vessel cavitation. Whale proxies use one to three localized events with variable duration, curvature, partials, taper, and flutter. Vessel-tonal and narrow-band cavitation components draw a 5–20 Hz low-frequency-oscillator amplitude envelope with probability 0.8; their broadband component is not modulated. Training SNR is uniform on $[-18, 12]$ dB. This distribution is intended to cover contour and modulation patterns, not to claim species identity.

We quantify spectral displacement with directional $D_{\mathrm{KL}}(p\|q)$ and finite symmetric Jensen–Shannon (JS) divergence [18]; spectral entropy and source–ocean overlap are reported separately. Larger KL/JS expands coverage but is not treated as semantic diversity. The DCLDE main experiment samples only its five whale and two vessel families; all six main-domain models see the same target distribution.

### D. Main data construction and secondary ablations

The main dataset uses day-disjoint 2-kHz DCLDE 2013 recordings and whale-call annotations [19], [20]. Across 168 h of WAV, excluding annotated calls and a 2-s guard on each side yields 146.4 h of annotation-clean 5-s windows. For each target, the 12 immediately preceding 5-s blocks from the same WAV form a high-resolution history. Annotated calls and their 2-s guards are masked; 235 of 68,075 eligible current windows have one masked historical block, filled by the nearest clean block still within the preceding 60 s. Histories never cross WAV boundaries or use the current window. This leaves 94.5 h eligible overall, split into 37.3/9.7/47.6 h for train/validation/test. For each current clean $N_t$, a seven-family whale/vessel synthetic event $s_t$ is mixed to form $x_t = N_t + s_t$. The main split is by recording day, and the GRU receives only event-masked features from the immediately preceding 5-s windows.

Secondary ablations use NOAA–Navy SanctSound hourly TOL products [21]. MB03 deployments 01–02/03/04 are train/validation/test. Random-phase inversion creates controlled spectral surrogates, not recovered audio. Concatenation, cross-attention, KL/JS coverage, and the one-hour history-resolution comparison are confined to this domain. The separate 42-h SanctSound WAV strength sweep also lacks event annotations, so it is a useful controlled real-background ablation but is not counted as clean-background primary evidence.

Complete-reference tiers report SI-SDR and SI-SDRi using scale-invariant projection [22], [23]. All uncertainty intervals are seeded paired bootstraps, blocked by recording day for the main test. The event classifier is frozen before any real-event evaluation. Current test features are used only as targets for the synthetic full-reference task, never as GRU input; all histories end before $t$.

## III. Experiments and Results

### A. Preliminary conditioning-mode ablation

The initial 30-band design selected FiLM. Table I compares it with alternative interfaces on controlled 5-s spectral-surrogate examples; on 832 fixed-0-dB pairs (64 per family), FiLM improves 4.850 to 5.541 dB SI-SDRi. Band-wise scale and bias directly modulate existing band evidence. Concatenation must compress doubled features, while cross-attention must learn alignment from coarse predicted spectra under the same budget.

FiLM gains 0.869, 0.835, 0.919, 0.906, and 0.806 dB on the five whale proxy families. Vessel tonal is the hardest family: the

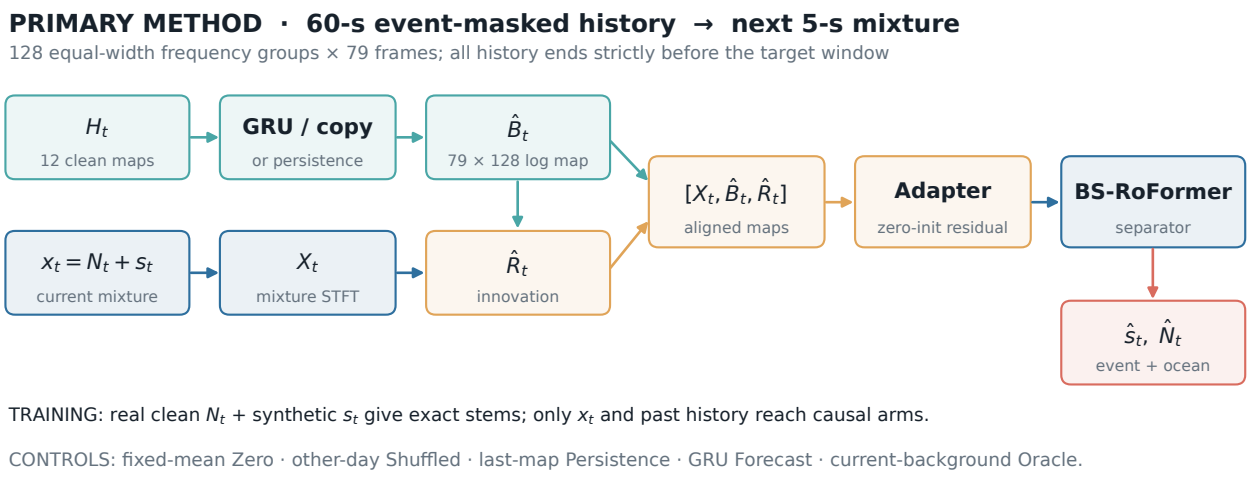



Fig. 1. Main 128-group causal residual architecture. The event-masked 60-s history predicts a full future 5-s background map; the explicit innovation is $\widehat{R}_t = \log|X_t| - \widehat{B}_t$. The same residual adapter is used for Zero, wrong-day Shuffled, Persistence, Forecast, and current-background Oracle conditions. Only the Oracle sees current clean background as an input.

TABLE I
CONDITIONING-MODE ABLATION ON TOL-DERIVED SPECTRAL SURROGATES (DB).

| Injection | Val. SI-SDRi | Test SI-SDRi | Paired $\Delta_0$ |
|---|---|---|---|
| None | 4.864 | 4.059 | 0 |
| Concatenation | 5.343 | 4.432 | +0.623 |
| FiLM | **5.419** | **4.468** | **+0.692** |
| Cross-attention | 5.128 | 4.261 | +0.331 |

conditioned absolute SI-SDRi is $-8.295$ dB, despite a 0.320-dB relative gain. Overall gain has weak correlation with JS ($r = -.057$) and entropy ($r = .122$), so increasing divergence broadens support but does not predict usefulness.

### B. Preliminary 30-band FiLM audit on DCLDE

The original FC-BSR receives twelve event-masked preceding 5-s features and a current clean 5-s DCLDE segment; the seven-family event is added only after the clean background is selected. The model therefore sees $x_t$ and $H_t$, while the exact $N_t$ and $s_t$ are supervision. The training split contains 37.3 h of eligible clean current audio. The 12-step GRU forecasts the next 5-s band spectrum; validation MAE is 1.926 dB versus 2.111 dB for persistence (8.8% lower). The older two-row comparison labeled a Zero-FiLM checkpoint as "BS-RoFormer." We retain its exact trained waveform mapping as the Zero control, attach the same frozen GRU in a bypassed branch, and train a true injector-bypassed BS-RoFormer. All six rows in Table II use the same mixtures, transferred backbone initialization, optimizer budget, validation selection, and 4,096 test examples from three held-out days. Figure 3 stratifies these same samples by input SNR.

The architecture effect, Zero minus BS-RoFormer, is $-0.0002$ dB (95% CI $[-0.0068, 0.0088]$). Forecast minus BS-RoFormer is $-0.134$ dB ($[-0.327, 0.125]$); Oracle minus BS-RoFormer is $-0.096$ dB ($[-0.238, 0.034]$). Oracle minus Forecast is only $+0.039$ dB ($[-0.090, 0.155]$). Shuffling and persistence also fail to show a positive average over Zero. Thus the matched controls do not establish that temporal forecast information improves this separator. Oracle's higher validation score (11.372 versus 10.946 dB for BS-RoFormer) does not transfer to a clear held-out gain. With only three test days, the intervals leave small effects unresolved; the 30-band oracle supplies more information, but separately optimized FiLM models do not form a guaranteed performance ordering.

*a) Full-waveform oracle copy check.:* To test whether losing information in $E$ explains the negative 30-band result, we freeze the same trained BS-RoFormer and fit a pointwise, three-coefficient head on 8,192 training mixtures:

$$\hat{s}_t = w_b \hat{s}_{\mathrm{BSR}}(x_t) + w_x x_t + w_c c_t. \tag{4}$$

The head starts at $(1, 0, 0)$, exactly reproducing BSR, and its waveform-MSE global optimum is solved by least squares. We fit the *same* head with zero, wrong-time, and actual current background waveforms; validation and testing use 4,096 examples each, with the test IDs, SNRs, mixture scores, and frozen BSR outputs checked against Table II. For $c_t = N_t$, the learned weights are $(0, 1, -1)$ to numerical precision: the head copies the exact answer $x_t - N_t = s_t$. The maximum test waveform error is $5.96 \times 10^{-8}$; the finite SI-SDRi of 80.755 dB is limited by the metric's numerical floor. Zero and wrong-time heads receive no useful background identity and do not improve SI-SDRi (Table III). This is an intentionally noncausal information/path sanity check with a different interface, not an architecture-matched FiLM ablation or evidence that the 30-band forecast can recover a phase-aligned background waveform.

*b) Oracle information ladder.:* We next reduce the information in the *current* $N_t$ while keeping the frozen BSR, the

TABLE II

PRELIMINARY 30-BAND FILM AUDIT ON 4,096 MATCHED ANNOTATION-CLEAN 5-S BACKGROUNDS PLUS SYNTHETIC EVENTS. VALUES ARE TARGET SI-SDRI (DB); BRACKETS ARE PAIRED 95% BOOTSTRAP INTERVALS OVER HELD-OUT RECORDING DAYS. THE 30-BAND ORACLE ALONE SEES $E(N_t)$.

| Model | Validation | Test | Δ vs. BSR [95% CI] | Δ vs. Zero [95% CI] |
|---|---|---|---|---|
| BS-RoFormer | 10.946 | 9.631 | 0 | $+0.0002$ $[-0.009, 0.007]$ |
| Zero-condition FiLM | 10.967 | 9.631 | $-0.0002$ $[-0.007, 0.009]$ | 0 |
| Shuffled-condition FiLM | 10.609 | 9.526 | $-0.105$ $[-0.247, 0.020]$ | $-0.105$ $[-0.256, 0.026]$ |
| Persistence-condition FiLM | 11.205 | 9.524 | $-0.107$ $[-0.239, 0.083]$ | $-0.107$ $[-0.232, 0.084]$ |
| GRU-Forecast FiLM (FC-BSR) | 11.213 | 9.497 | $-0.134$ $[-0.327, 0.125]$ | $-0.134$ $[-0.320, 0.125]$ |
| 30-band Oracle-FiLM, $E(N_t)$ | 11.372 | 9.535 | $-0.096$ $[-0.238, 0.034]$ | $-0.095$ $[-0.231, 0.035]$ |

TABLE III

FULL-WAVEFORM ORACLE COPY CHECK ON THE SAME 4,096 DCLDE TEST MIXTURES. TARGET SI-SDRI AND PAIRED DAY-BLOCK Δ AGAINST THE FROZEN BSR (DB).

| Frozen BSR + fitted head | SI-SDRi | Δ [95% CI] |
|---|---|---|
| BSR; initial $(1, 0, 0)$ | 9.631 | 0 |
| Zero waveform | 9.231 | $-0.400$ $[-0.483, -0.269]$ |
| Wrong-time waveform | 9.231 | $-0.400$ $[-0.483, -0.269]$ |
| Actual $N_t$ waveform | 80.755 | $+71.124$ $[69.945, 72.081]$ |
| Analytic $x_t - N_t$ | 80.755 | $+71.124$ $[69.945, 72.081]$ |

three-coefficient head, training examples, and 4,096 matched test mixtures fixed. A 512-point Hann STFT at 2 kHz has 257 frequency bins on a 64-ms hop. We average background power across either adjacent equal-width groups (128, 64, 32) or geometric 20–950-Hz groups (128, 64, 48, 30), and separately pool frames over 0.25, 1, or 5 s. The retained magnitude is inverted with either the actual $N_t$ phase or the mixture $x_t$ phase. Geometric groups absorb edge bins; only 95 of the nominal 128 geometric groups contain an STFT bin. The 30-band/5-s rung is analogous to, but not numerically identical with, the original $E(N_t)$.

Figure 2 identifies where this *copy path* ceases to show a held-out gain. Exact amplitude with mixture phase still adds 11.164 dB over BSR (day-block CI $[10.889, 11.594]$). With equal-width groups, 128 retains $+2.067$ dB ($[0.969, 3.386]$), while 64 has $-0.623$ dB ($[-1.404, 0.259]$): the confirmed gain disappears between these levels. For geometric bands, nominal 128 already gives $-0.529$ dB ($[-1.257, 0.319]$), and 30 per-frame bands give $-0.672$ dB ($[-1.323, 0.086]$). Retaining the true phase at 30 per-frame bands gives only $+0.117$ dB ($[-0.671, 1.084]$). By contrast, keeping all 257 bins but pooling power to 1 s still yields $+0.585$ dB with mixture phase ($[0.285, 1.132]$); the 5-s version yields $+0.249$ dB ($[-0.044, 0.495]$). Thus this interface is especially sensitive to frequency averaging and the grouping rule, while time averaging alone is less destructive. These are sufficiency results for the fixed waveform copy head, not information-theoretic limits or new FiLM results.

### C. Main 128-group residual experiment

*a) Trained high-resolution Oracle residual separator.:* We next test the representation in an actual BS-RoFormer, rather than only a frozen linear copy head. The actual current $N_t$ and mixture $x_t$ are transformed with a 512-point, 128-hop STFT. Mean power in 128 equal-width frequency groups yields time-resolved $\log|B_t|$; the explicit innovation is $R_t = \log|X_t| - \log|B_t|$ on the same grid. These maps are interpolated to the BSR frame grid and projected into its mixture-derived band tokens before axial attention. This zero-initialized residualized $[X_t, B_t, R_t]$ input uses neither FiLM nor a forecaster; its initial waveform output exactly matches ordinary BSR (maximum difference zero).

Both models use the same day-disjoint DCLDE mixtures, transferred backbone initialization, seed, batch order, 6,000 updates, AdamW/cosine schedule, loss, and validation-best checkpoint rule. The existing BSR selected step 5,300; the Oracle model selected step 6,000. On 4,096 matched test examples, BSR scores 9.631 and Oracle-Residual-BSR 16.767 dB target SI-SDRi. The paired improvement is $+7.136$ dB (recording-day 95% interval $[6.970, 7.282]$), with all three held-out days positive. The combined high-resolution prior and explicit innovation interface therefore benefits this controlled separator. Actual $N_t$ is unavailable at deployment; this two-model comparison also does not isolate adapter capacity from the condition information.

*b) Causal high-resolution residual conditions.:* We then replace the current-background Oracle map with two conditions computed strictly from the preceding 60 s. Each of twelve 5-s history windows has the same $79 \times 128$ log-magnitude grid. Annotation-overlapping windows are masked; when one is unavailable, its slot copies the nearest annotation-clean window still inside the past 60 s. This affects 235 of 68,075 eligible current windows. Persistence uses the last clean history map. A lightweight GRU encodes each map into four temporal chunks, processes twelve steps, and decodes a residual correction to persistence. It predicts the entire future $79 \times 128$ log-background map, without seeing the current waveform. The predictor is trained for 3,000 updates on training-day clean backgrounds and frozen at its lowest validation log-magnitude MAE. Validation MAE is 0.5456 for the GRU versus 0.5518 for persistence (1.12% relative gain). On the held-out test days, the corresponding map MAEs are 0.5308 and 0.5386.

For every condition map $B$, we form $R = \log|X_t| - B$ and feed $[X_t, B, R]$ through precisely the Oracle residual adapter. The Zero control repeats the fixed training-set mean log background at every frame. The Shuffled control draws an annotation-clean $B_j$ from a *different recording day*: training and

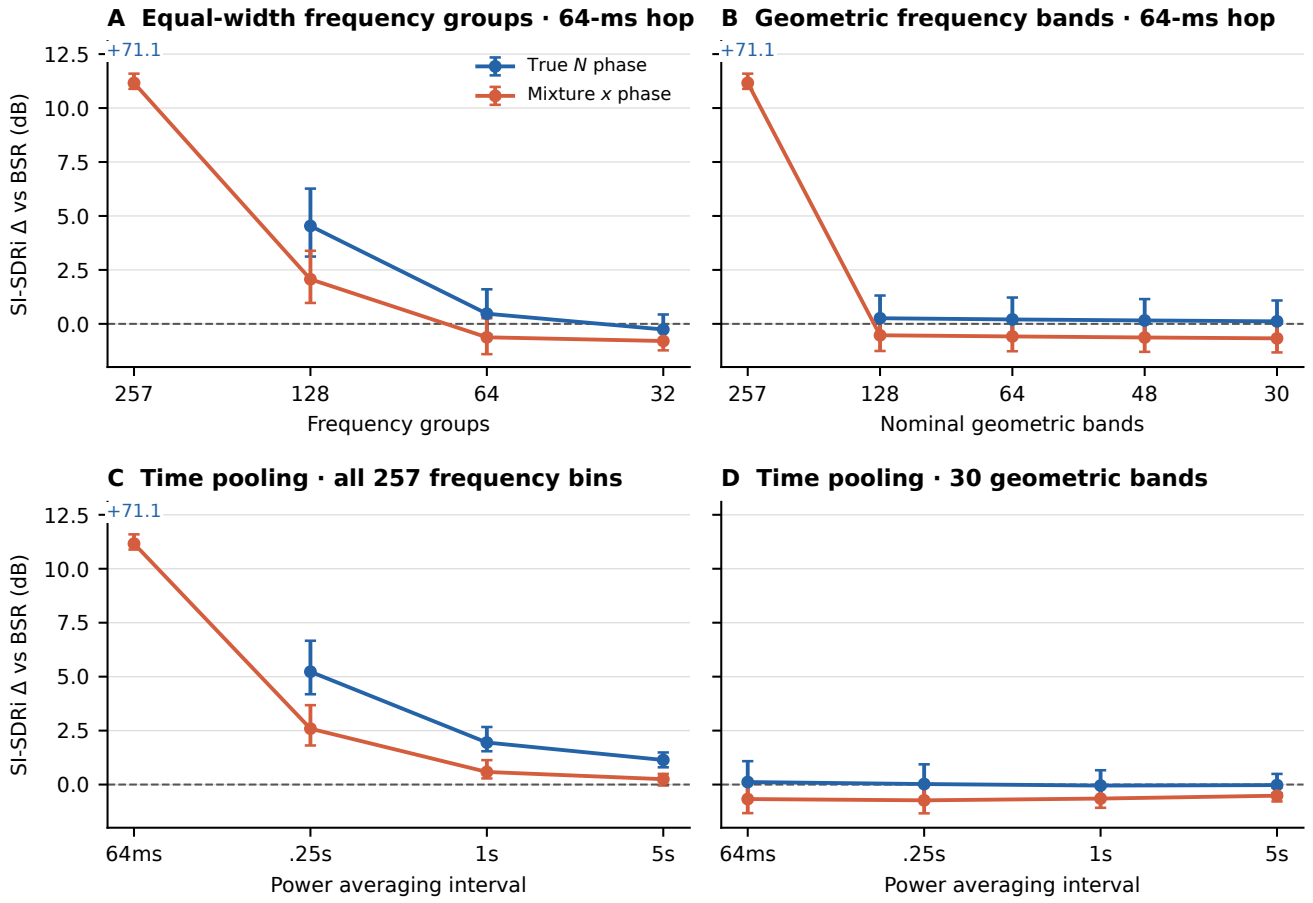



Fig. 2. Current-background Oracle information ladder on identical DCLDE test mixtures. Each point fits the same three-coefficient copy head on training mixtures. Vertical bars are paired 95% day-block intervals for SI-SDRi difference from frozen BSR. Full-resolution true-phase $N_t$ gives +71.1 dB and is above the plotted range; mixture phase is a separate information loss. The threshold depends on frequency grouping and this copy interface.

testing draw within their respective splits; validation draws from training days because validation has only one day. Persistence takes the last clean history map; Forecast uses the frozen GRU. All four separators start from the same transferred BSR backbone and zero adapter, so their initial waveform output is identical to ordinary BSR. They use identical synthetic events, training-example order, 6,000 update budget, optimizer, loss, validation-best checkpoint rule, and held-out mixtures. Thus the four residual arms differ only in the source of $B$.

Table IV separates interface and background effects. Zero exceeds ordinary BSR by $+2.525$ dB (recording-day paired 95% CI $[2.262, 2.701]$); Shuffled exceeds BSR by $+2.313$ dB ($[2.027, 2.905]$). Hence the full BSR-to-causal gain cannot be attributed to correct history alone. The Zero contrast includes both the extra adapter and the current mixture log-spectrum supplied by the innovation path, so it does not separately identify parameter count. With the adapter held fixed, Persistence exceeds Zero by $+2.138$ dB ($[1.775, 2.377]$) and wrong-day Shuffled by $+2.351$ dB ($[1.571, 2.661]$), positive on every held-out day. This supports a contribution from temporally relevant historical background beyond architecture and arbitrary background maps. Forecast exceeds Persistence by only $+0.056$ dB ($[0.036, 0.074]$), again positive on all three days but small in practice. Forecast retains 66.1% of the Oracle gain over BSR, with a 2.416 dB Oracle gap ($[2.114, 2.640]$).

TABLE IV
MATCHED DCLDE TEST ON 4,096 CLEAN-BACKGROUND MIXTURES. ALL FIVE RESIDUAL ARMS USE THE IDENTICAL 128-GROUP ADAPTER; ZERO USES A FIXED TRAINING MEAN MAP AND SHUFFLED USES ANOTHER RECORDING DAY. ONLY ORACLE RECEIVES THE CURRENT TRUE BACKGROUND. VALUES ARE MEAN TARGET SI-SDRI IN DB.

| Condition | SI-SDRi | Gain vs. BSR |
|---|---|---|
| BSR (mixture only) | 9.631 | — |
| Zero-Residual | 12.156 | +2.525 |
| Shuffled-Residual | 11.944 | +2.313 |
| Persistence-Residual | 14.295 | +4.664 |
| GRU-Forecast-Residual | 14.351 | +4.720 |
| Oracle-Residual | 16.767 | +7.136 |

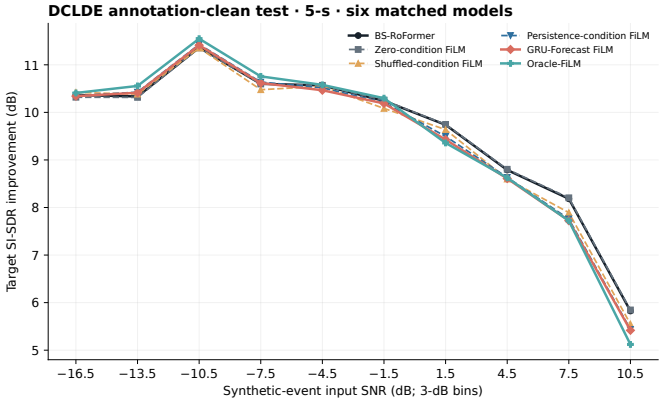



Fig. 3. Preliminary 30-band FiLM audit on annotation-clean DCLDE backgrounds: mean target SI-SDRi in 3-dB input-SNR bins for all six matched methods. Table II gives day-block uncertainty for their overall means.

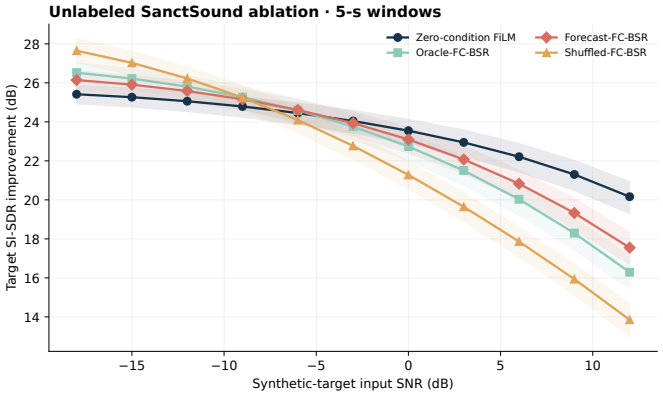



Fig. 4. Ablation on 5-s unlabeled SanctSound background: target-strength sweep. Lines show the four conditions; shaded bands are paired 95% intervals.

TABLE V
ABLATION: PREDICTOR-RESOLUTION DIAGNOSTIC ON SANCTSOUND BACKGROUND.

| Step | Predictor | Cond. | Base | Gain [95% CI] |
|---|---|---|---|---|
| 5 s | GRU | 29.753 | 28.527 | 1.226 [1.082,1.381] |
| 30 s | Persist. | 29.630 | 28.534 | 1.095 [0.935,1.265] |
| 1 min | Persist. | 29.661 | 28.533 | 1.128 [0.971,1.298] |
| 10 min | Persist. | 29.711 | 28.535 | 1.176 [1.036,1.326] |

Cond./Base/Gain are target SI-SDRi in dB.

### D. Secondary ablations: time scale and target strength

The time-resolution diagnostic in Table V is an ablation on continuous SanctSound audio, not the main annotation-clean data. Only its 5-s predictor beats persistence on validation (MAE 1.366 versus 1.386 dB); the other scales select persistence. The separate 42-h strength curve also uses unlabeled current background. It shows that forecast conditioning is strength-dependent in that domain and must not be read as a result on annotation-clean current segments.

The harder fixed-5-s experiment on unannotated SanctSound WAV compares Zero-condition FiLM, a current-spectrum oracle, causal forecast, and a time-shuffled condition on 1,152 paired test mixtures. This oracle belongs to the secondary data domain; Table II uses the actual annotation-clean DCLDE $N_t$. Averaged across training SNRs, the baseline scores 23.470 dB and Forecast-FC-BSR 22.871 dB ($\Delta = -0.599$, CI $[-0.851, -0.346]$); even the oracle is lower. This exposes conditioning/capacity effects that the simpler diagnostic misses. Figure 4 gives the more informative strength sweep. Forecast conditioning helps weak targets by 0.728, 0.643, 0.525, 0.366, and 0.157 dB at $-18, -15, -12, -9, -6$ dB, respectively, then loses from $-3$ dB onward. The shuffled condition also helps the weakest targets but collapses as SNR rises; therefore weak-target gain alone cannot prove that temporal forecast accuracy, rather than regularization or extra capacity, is responsible.

### E. Real events without separated ground truth

Following learned reference-free metrics [24], [25] and semantic audio embeddings [26], [27], we train, calibrate, and freeze a binary event classifier (test AUROC .9723, AP .9755, F1 .9275). Mixture consistency imposes $\hat{N} = x - \hat{s}$. We restore $\hat{s}$ at gains $\{-60, -30, -24, -18, -12, -6, 0\}$ dB and test whether event probability recovers monotonically. Identity and all-to-event controls reject trivial solutions; changes outside annotated intervals audit preservation.

We evaluate the four frozen 128-group residual models on the exact 103 test calls used by the preliminary FiLM audit. Each call receives twelve strictly earlier annotation-clean maps from the same WAV. Six histories each lack one clean window, filled by the nearest clean window still inside the prior 60 s. Zero uses the training mean, Shuffled a clean background from another test day, Persistence the last historical map, and Forecast the frozen GRU applied to that history. No condition reads the current call. Table VI and Fig. 6 show the results.

Persistence versus Zero lowers $p(\hat{N})$ by .0624 (recording-day paired 95% CI $[-.0682, -.0547]$), raises $p(\hat{s})$ by .0469 ($[.0198, .1006]$), and lowers off-event change by .0255 ($[-.0386, -.0077]$). Forecast shows the same directions versus Zero, with $\Delta p(\hat{N}) = -.0531$ and $\Delta$ off-event $= -.0260$. However, wrong-day Shuffled attains lower $p(\hat{N})$ and higher $p(\hat{s})$ than either causal condition: Persistence minus Shuffled is $+.0471$ ($[.0317, .0706]$) for $p(\hat{N})$ and $-.0606$ ($[-.0731, -.0404]$) for $p(\hat{s})$. Persistence protects off-event audio much better ($\Delta = -.0408$, $[-.0519, -.0320]$). Monotonicity is near saturation for all four models and does not separate them. The classifier probabilities therefore do not reproduce the controlled full-reference ranking; they can reward aggressive extraction, while off-event preservation favors the causal conditions. Isolated real-event stems remain unavailable.

### F. Classifier floor and synthetic-target coverage

To audit the high $p(\hat{N})$, we apply the frozen classifier to annotation-clean test audio, truncated to the separator's 9,984 output samples. Its mean probability is .153 on 4,096 day-proportional clean windows and .176 on 103 clean windows

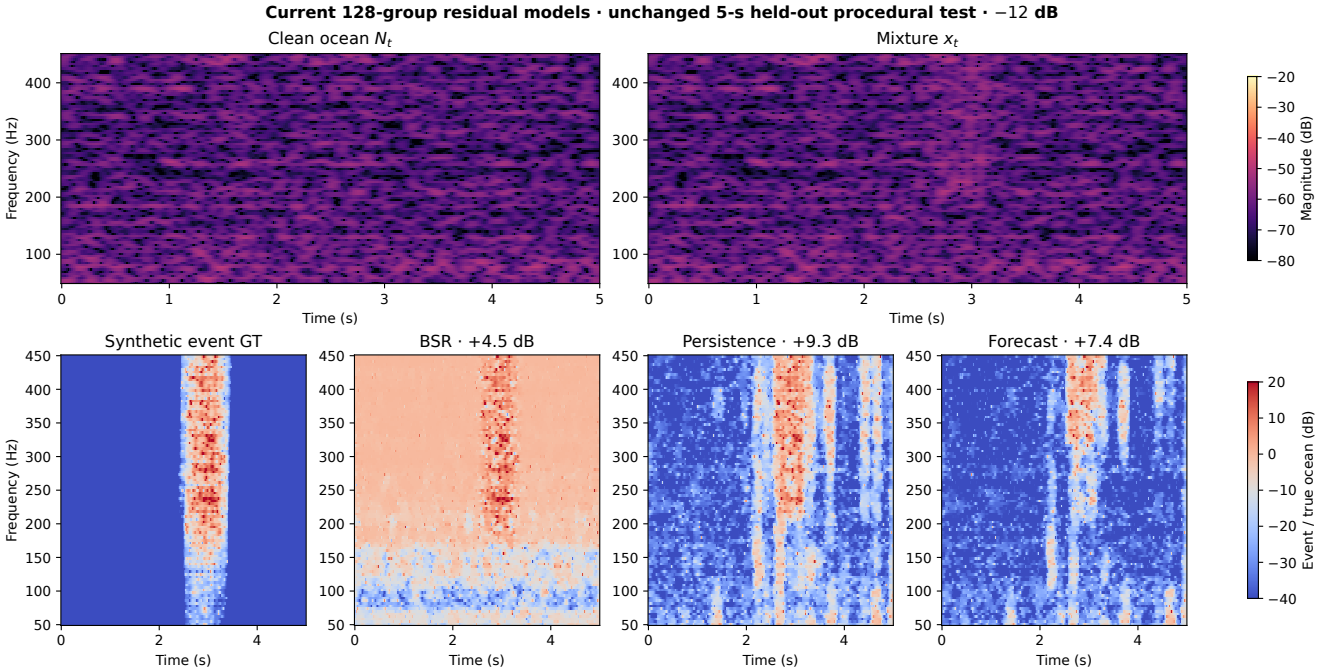



Fig. 5. Current 128-group residual separators on an unchanged held-out $-12$ dB procedural whale pulse packet, selected by joint median SI-SDRi among 96 fixed examples. The ocean and mixture share an absolute magnitude scale; event stems share a dB scale relative to the true ocean. Compared with BSR, causal Persistence and Forecast suppress much of the off-target ocean, though isolated vertical artifacts remain. Across all 96 examples, the median event/ocean contrast in ground-truth-absent TF bins is $-1.37$, $-31.44$, and $-36.46$ dB for BSR, Persistence, and Forecast. These contrasts measure controlled synthetic leakage, not real-whale stem fidelity.

TABLE VI
FROZEN 128-GROUP RESIDUAL MODELS ON THE SAME 103 REAL CALLS. ALL MODELS SEE THE SAME MIXTURE; REAL SEPARATED STEMS ARE UNAVAILABLE. OFF-EVENT IS RELATIVE BACKGROUND CHANGE OUTSIDE ANNOTATED CALLS.

| Method | $p(x)$ | $p(\hat{N})\downarrow$ | $p(\hat{s})\uparrow$ | Mono.↑ | Off-event↓ |
|---|---|---|---|---|---|
| Zero-Residual | .909 | .891 | .802 | .990 | .062 |
| Shuffled-Residual | .909 | .781 | .910 | .989 | .078 |
| Persistence-Residual | .909 | .829 | .849 | .992 | .037 |
| GRU-Forecast-Residual | .909 | .838 | .842 | .989 | .036 |

from the same WAV, immediately before the corresponding real calls; both medians are .093. The real mixtures average .909. The four separated-background means of .781–.891 are well above the clean floor; paired differences from the same-WAV control range from .606 to .715. This is a classifier-response gap, not a waveform measure of leaked whale energy: artifacts or unmarked events can also elicit responses.

We generate 32 whale-target mixtures per matched clean window (3,296 total), using the five training whale families and $[-18, 12]$ dB sampled strengths. Their mean classifier probability is .407. For a proxy coverage test, each event-bearing clip's frozen 64-D penultimate activation is subtracted from its same-WAV clean activation. After scaling dimensions by synthetic variability, median real-to-real leave-one-out nearest-neighbor distance is .233. Searching all 3,296 synthetic clips gives a real-to-synthetic median of .262, but this reference set is much larger. Using one synthetic clip per real call gives 103 candidates per repeat; across 32 repeats the median distance is .483, or $2.08\times$ the real-to-real distance. A median $51.5\%$ of calls exceed the real-to-real 95th-percentile distance threshold (versus $3.9\%$ with all 3,296 synthetic candidates). Synthetic examples are thus sparse near real events in this *classifier-dependent, equal-size* representation. This motivates target-distribution refinement; neither the embedding nor detector confidence establishes the physical cause of residual event responses without isolated real whale stems. Real background evolution since the preceding clean window also enters this embedding difference.

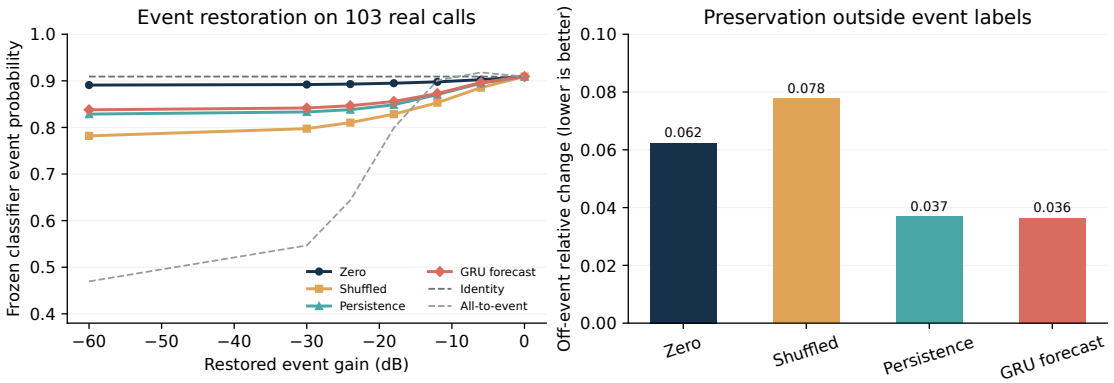



Fig. 6. Frozen-classifier event restoration and off-event preservation for four matched residual models. Identity and all-to-event are degenerate controls. Shuffled scores strongly on event probabilities but changes more audio outside annotated calls than either causal condition.

### G. Training-day target calibration and acceptance

We next use only the three training dates to fit target morphology. Of 2,798 annotated call windows, 2,616 have usable same-WAV clean windows before and after the call and sufficient acoustic excess. We interpolate their local background log spectra, take positive mixture-minus-background excess in the annotated time span and 40–950 Hz, apply a smooth mask, and invert with mixture phase to obtain a *pseudo* whale stem. This is an approximate target, not an isolated reference. We estimate call duration, frequency limits and track, sweep slope and curvature, bandwidth, harmonic count and decay, amplitude envelope, jitter, pulse gap, and event count from these training calls. A calibrated procedural route perturbs a sampled whale profile while retaining the original tonal and cavitation vessel generators; the second route randomizes timing and phase of a pseudo-real stem. Each contributes half the new target pool, with vessel draws occupying $4/7$ of the procedural half to preserve the original overall whale/vessel ratio in expectation. The held-out classifier is not used to fit profiles or filter samples.

Under the same 103-call, 32-repeat equal-size coverage protocol, with both new routes represented in every repeat, the median real-to-target nearest distance falls from .483 to .322; the median fraction above the real-to-real 95th-percentile threshold falls from $51.5\%$ to $3.4\%$. Distance improves in all 32 paired repeats, meeting our pretraining acceptance criterion. The frozen classifier's mean response to new mixtures rises from .407 to .647, but this is an external diagnostic, not an optimization target or proof of stem fidelity.

### H. Separation after target-pool revision

We retrain Persistence and Forecast residual separators for 6,000 updates each, with the same transfer initialization and validation selection; the background GRU and event classifier remain frozen. Table VII compares them with their original counterparts. On the same real calls, Persistence lowers $p(\hat{N})$ by .1766 (recording-day paired 95% CI $[-.1920, -.1479]$) and raises $p(\hat{s})$ by .0668 ($[.0414, .1166]$). Forecast changes these probabilities by $-.1129$ and $+.0527$. Both gains coexist with greater off-event change: $+.0215$ for Persistence ($[.0151, .0324]$) and $+.0068$ for Forecast ($[.0002, .0132]$). New Persistence extracts more event-like audio than new Forecast, but changes more audio outside the annotation. On the unchanged legacy synthetic test, SI-SDRi falls by $2.100$ and $2.620$ dB, respectively. The frozen GRU does not improve over persistence after target-pool revision. Retaining vessel draws prevents the severe tonal-vessel regression of an intermediate whale-only pool, yet does not preserve the old whale-proxy scores. Even the new $p(\hat{N})$ values remain far above the matched clean floor of .176. Without isolated real stems, these findings support a target-domain effect on classifier response, not a full-reference real separation gain.

To check optimization separately, we train fixed small sets. With the same loss and backbone, 128 old procedural mixtures rise from $13.85$ to $17.20$ dB train SI-SDRi in 3,000 updates; 128 empirical mixtures rise from $10.57$ to $15.56$ dB in 20,000 updates, with little further SI-SDRi gain near the end. Their target-absent event/ocean contrast nevertheless falls from $-32.85$ to $-60.68$ dB. This distinguishes remaining target reconstruction error from simple background leakage on these controlled mixtures. We then fine-tune the original strong Persistence checkpoint through $0\%$, $25\%$, and $50\%$ pseudo-real stages at one-tenth the original learning rate. Disjoint 80/20 training-day pseudo profiles and fixed old/empirical validation suites select step 6,500 in the $25\%$ stage: validation SI-SDRi changes from $16.26/11.12$ to $16.44/11.84$ dB. Four additional thousand $50\%$-stage updates further suppress target-absent energy but reduce joint SI-SDRi. The selected checkpoint scores $14.421$ dB on the untouched old-synthetic test, yet its real-event $p(\hat{N})/p(\hat{s}) = .819/.823$ fails to retain the $50/50$ model's $.652/.916$. Thus more training helps a fixed set and avoids old-domain forgetting, but alone does not resolve the real-target tradeoff. A matched SI-SDR auxiliary term gains $.40$ dB on a fixed 128-mixture training probe, yet four thousand full-data $50\%$-stage updates do not exceed the same dual-validation starting point. We therefore keep the original loss; adding further target-absent suppression is also unsupported by the current controlled leakage audit.

TABLE VII
SAME FROZEN REAL-CALL EVALUATION AFTER TARGET-POOL REVISION, AND SI-SDRI ON THE UNCHANGED OLD SYNTHETIC TEST. CURRICULUM SELECTS ON TWO SEPARATE VALIDATION SUITES; THE 128-GROUP INTERFACE AND PREDICTOR ARE FIXED.

| Condition | Training targets | Old-synth SI-SDRi (dB) | Real $p(\hat{N})\downarrow$ | Real $p(\hat{s})\uparrow$ | Off-event↓ |
|---|---|---|---|---|---|
| Persistence | Original procedural | 14.295 | .829 | .849 | .037 |
| Persistence | 50/50 empirical | 12.195 | .652 | .916 | .058 |
| Persistence | 50/50 + real weak | 10.323 | .408 | .978 | .061 |
| Persistence | Frozen trunk + clean-match teacher | 12.156 | .461 | .960 | .064 |
| Persistence | Curriculum (25% pseudo) | 14.421 | .819 | .823 | .037 |
| GRU Forecast | Original procedural | 14.351 | .838 | .842 | .036 |
| GRU Forecast | 50/50 empirical | 11.731 | .725 | .895 | .043 |

### *I. Weak supervision from annotated real calls*

We next fine-tune the empirical Persistence separator with two train-day synthetic/pseudo-real mixtures and two train-day annotated real calls per update. Only 180 real calls have eligible clean causal histories; 47 validation-day calls select the checkpoint, and the 103 test calls remain untouched until selection. A separate temporal dilated CNN, trained from train-day onset/offset masks, supplies frozen framewise logits $C_A$; its validation-day frame average precision is .901. It differs in both architecture and training objective from the frozen clip-level test classifier $C_B$. Neither $C_B$ nor any test-day call enters fine-tuning or checkpoint selection.

For real mixture $x$, we set $\hat{N} = x - \hat{s}$ exactly and penalize $C_A(\hat{N})$ and $1 - C_A(\hat{s})$ within annotated frames. Outside the annotation we penalize the event-stem RMS relative to the mixture RMS. The supervised half retains the original multiresolution spectral loss. We give the weak term one-tenth weight because the unscaled detector loss rapidly degraded validation SI-SDRi. Selection minimizes validation-day background response subject to event response, off-event alteration, and old-synthetic SI-SDRi constraints fixed before opening the test set. The selected step is 250 of 3,000; later updates fail the SI-SDR constraint.

On the independent $C_B$ and the same 103 calls, $p(\hat{N})$ changes from .652 to .408 (recording-day paired 95% CI for the difference $[-.365, -.093]$) and $p(\hat{s})$ from .916 to .978 ($[.044, .072]$). Off-event relative change is .058 versus .061; its paired CI $[-.004, .008]$ includes zero. The mean matched-clean probability is .176. The per-call clean-normalized logit residue $\operatorname{logit} p(\hat{N}) - \operatorname{logit} p(N_{\text{clean,matched}})$ falls from 3.606 to 1.389. On the unchanged 4,096-mixture synthetic test, however, SI-SDRi falls from 12.195 to 10.323 dB. These independent-detector results support partial transfer from real event annotations, but do not reach the clean-background response or establish waveform-level recovery of an unobserved real stem.

To limit forgetting, we freeze all band encoding and attention layers and train only the residual projection and mask output head. The original 50/50 Persistence model is a frozen teacher. The new objective retains supervised synthetic/pseudo-real loss, adds two-stem $L_1$ teacher consistency, and matches $C_A(\hat{N})$ to $C_A$ on the immediately preceding annotation-clean 5-s window from the same WAV. Event retention and off-event preservation remain in the real loss. Specifically, $L = L_{\text{GT}} + .01L_{\text{real}} + .1L_{\text{teacher}}$, where $L_{\text{real}} = .25L_{\text{match}} + .1L_{\text{event}} + 2L_{\text{off}}$; $L_{\text{match}}$ is a frame-masked smooth-$L_1$ loss on training-detector logits. Of 414,658 parameters, 99,774 representation parameters stay frozen; all 241 frozen state tensors are bit-identical to the teacher in the selected checkpoint. Validation-day constraints select step 900. A stronger teacher-weight trial does not materially improve the validation clean-response gap, so it is not promoted to test.

On the reused 103-call test cohort, independent $C_B$ gives $p(\hat{N})/p(\hat{s}) = .461/.960$, versus $.652/.916$ before weak tuning and $.408/.978$ after full-network weak tuning. Off-event change is .064, versus .058 before tuning; its paired increase is positive across the three dates. Crucially, unchanged old-synthetic SI-SDRi is 12.156 dB, only .038 dB below the 50/50 teacher and 1.833 dB above full-network weak tuning. The clean-normalized logit residue is 1.917 (teacher: 3.606; full weak: 1.389). This is a more conservative Pareto point, but it does not reach .25–.30 background event probability or the matched-clean response of .176. Because this cohort was already reported for the preceding model, this follow-up comparison is exploratory rather than a fresh blind test.

## IV. DISCUSSION AND CONCLUSION

The preliminary 30-band FiLM model does not gain even from the true current 30-band background. A successful full-waveform copy check and the fixed-head information ladder show why this negative result cannot rule out useful background information: compression and injection remain confounded. With a 128-group residual interface, Oracle gains 7.136 dB over BSR; past-60-s Persistence and Forecast retain 4.664 and 4.720 dB. Zero and wrong-day Shuffled also gain 2.525 and 2.313 dB, so the expanded interface contributes. Within that interface, correct history adds 2.138 dB over Zero and 2.351 dB over Shuffled. The GRU adds only .056 dB over persistence on the original target distribution, with 2.416 dB remaining to Oracle.

On real calls, the original four-condition ranking differs between frozen classifier probabilities and off-event preservation. The clean-background floor does not explain the high $p(\hat{N})$, and an equal-size embedding audit finds sparse synthetic

coverage near real calls. Training-day empirical targets close much of that measured gap and improve both real-call event probabilities, while increasing off-event changes and lowering old-domain SI-SDRi. The frozen GRU does not beat persistence after this target revision. Independent-detector weak fine-tuning lowers the background event proxy further while preserving the event-stem proxy and approximately maintaining off-event audio, but costs another 1.872 dB on the unchanged synthetic test. Freezing the separator representation and adding teacher consistency nearly eliminates that controlled-domain loss, at the expense of weaker real event suppression and a small off-event increase. Thus proxy improvement does not establish a uniform real separation gain. Reuse of three test days, uncertain event labels, and unavailable isolated stems further limit inference.

The existing 30-band FiLM frontend permits inspection and playback, but the new 128-group model has not been deployed there. Future work should preserve background during target adaptation, test more sites and target pools, and obtain stronger real-event references. Classifier restoration is a semantic proxy, not a waveform-fidelity measure.

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